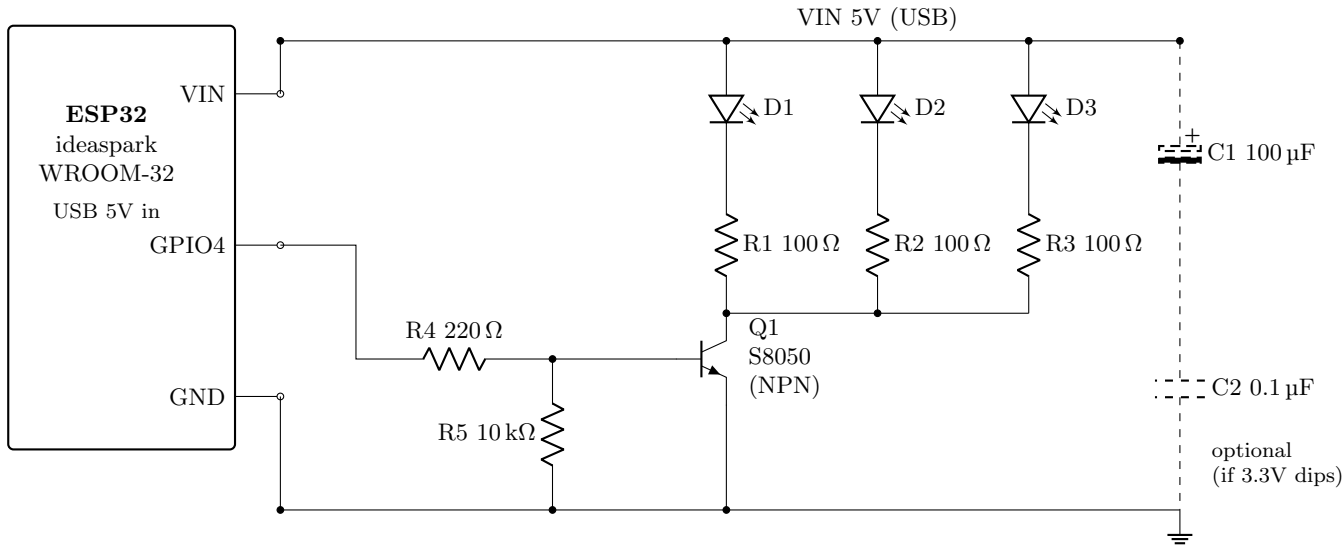




D1–D3: 940 nm IR LED (clear), 100 mA continuous or better

R1–R3: $I \approx (5.0 - 1.2 - 0.2)/100 \approx 36$ mA per branch, ≈ 108 mA total (calc.; measure). Use ≥ 0.25 W. Try 68 Ω only after measuring.

R4: $I_B \approx (3.3 - 0.7)/220 \approx 11.8$ mA (GPIO default drive 20 mA). Forced $\beta \approx 9$, Q1 saturated.

R5: keeps base LOW during reset / boot. Power: USB charger 1 A or more, not a PC port.

All grounds common (ESP32 GND, VIN return, Q1 emitter).